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Comparative Quantitative Study of the Hippocampal Region of Two Closely Related Species of Wild Mice: Interspecific and Intraspecific Variations in Volumes of Hippocampal Components

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ABSTRACT

To investigate the structural changes in the hippocampal region (subiculum, Ammon's horn, and area dentata) associated with speciation, the volumes of homologous components of this region of the forebrain were compared in the two closely related murine rodent species, *Apodemus flavicollis* (yellow-necked wood mice) and *A. sylvaticus* (long-tailed field mice), and in two geographically separated groups of *A. sylvaticus*.

With the exception of the mossy fiber zone, no significant differences were found in the relative sizes of the components of Ammon's horn. Significant interspecific differences were found in the deep subiculum, in the intermediate (medial perforant path) and deep (commissural-associational) zones of the molecular layer of the area dentata, and in the hilus. A significant intraspecific difference was found in the combined intermediate and superficial (medial and lateral perforant path) zone of the dentate molecular layer.

Differences found in the relative size of the hilus and the mossy fiber zone of Ammon's horn were complementary in that the combined volumes of these zones, which are both terminal fields of dentate granule cells, did not differ in the two species. This finding provides evidence that the distribution of the terminal field of a projection system can be altered while the size of the terminal field is maintained.

Within the hippocampal region, components of Ammon's horn appear well suited for quantitative comparative studies that span taxonomic units beyond the species level. In agreement with previous quantitative studies, differences in the volumes of components of Ammon's horn found in species of different orders are more likely to reflect phylogenetic trends rather than changes resulting from specializations of the particular species used. This is not true for the subiculum and the components of the area dentata.

Experimentally induced changes in the connectivity of the hippocampal region are discussed in terms of the structural changes which may be responsible for the quantitative differences observed between the two species studied here.

Key words: Apodemus spp., hippocampus, area dentata, subiculum, morphometry, species differences

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Three factors have a major influence on the structural organization of the brain of an individual: first, the phylogenetic history of the larger taxonomic unit to which it belongs, that is, the family or the order; second, the phylogenetic change associated with speciation or the development of strains, which subserves unique demands on the information processing of a group of individuals; and third, the ontogenetic history of the individual. When the phylogenetic development of a particular region of the brain or its phylogenetic state within a species is studied, the volume of this division of the brain is a commonly used parameter. In recent years, a considerable amount of information about the volumes and weights of different parts of the brains of numerous species has been published (recent compilations: Jolicoeur et al., '84; Kruska, '87; Stephan et al., '81; Stephan and Dieterlen, '82). These data reveal trends in the phylogenetic development of the brain, such as the allometrically progressive development of the limbic structures from insectivores to primates (Stephan, '83), as well as specializations that could be related to particular patterns of life, such as those seen in the neocortices of frugivorous bats and semiaquatic insectivores, which are larger than those of insectivorous bats and terrestrial insectivores (Jolicoeur et al., '84; Stephan and Kuhn, '82; Stephan et al., '74).

While the above mentioned studies have been concerned primarily with the major subdivisions of the brain, a number of recent studies have focused on more localized structures, such as the lateral geniculate body (Brauer et al., '82), the vestibular nuclei (Matano, '86), and the terminal fields of particular fiber systems in the hippocampal region (West and Andersen, '80; West et al., '84, West and Schwerdtfeger, '85). Studies of the components of the visual systems of closely related species have demonstrated that a considerable amount of variation can be expected at these levels of structural organization (Ebbeson, '80; Ulinski, '80). Although the development of species differences is the substrate of further phylogenetic development, these differences are not necessarily propagated in phylogeny and may be superimposed on other different phylogenetic trends. In order to reveal trends in the phylogenetic development of more localized structural entities of the brain, it is consequently necessary to determine the degree to which the volumes of these entities can vary in closely related species (Finlay et al., '87). To avoid the confounding effects of interspecific variations within a narrow taxonomic scale, comparisons of species belonging to different families or orders should include more species and should be confined to parts of the brain in which there is a little variation in closely related species. Similarly, comparisons of closely related species should be based on parts of the brain within which there is only a limited intraspecific variation. A description of the interspecific and intraspecific variations in volumes of the components of the hippocampal region of two closely related species, as intended in this study, provides a minimum estimate of the variation that can be expected from examination of the components of the hippocampal region.

The murine rodents *Apodemus flavicollis* (yellow-necked wood mice) and *A. sylvaticus* (long-tailed field mice) are two very closely related species of wood mice. In large parts of their common range, it is difficult to distinguish between them on the basis of their outer morphology. Prior to the work of Niethammer ('69) and Engel et al. ('73), their status as true noninterbreeding species was debatable. Since then, their status as true species has been confirmed by a large number of behavioral (Hoffmeyer, '73; Montgomery, '78), ecological (Montgomery, '80a,b), and protein taxonomic studies (Fraguedakis-Tsolis et al., '83; Gemmeke, '80). The criteria used to identify doubtful specimens are based primarily on measurements made on the skull, suggesting that the organization of the cranial contents may differ significantly between species. In addition, the extensive distribution of these species in Europe provides an opportunity to study groups within a species that are spatially separated and can be assumed to be adapted to different ecologic conditions. To our knowledge, a similar pair of closely related species is not found among commonly used laboratory animals.

The hippocampal region of the brains of rodents is particularly well suited for quantitative morphologic studies. The main neuron populations, the pyramidal cells of Ammon's horn and the dentate granule cells, are confined to distinct layers; the major extrinsic and intrinsic afferents, like the perforant pathways (Hjorth-Simonsen and Jeune, '73; Steward, '76), the commissural-associational systems (Blackstad, '56; Hjorth-Simonsen and Laurberg, '77; Swanson et al., '78) and the mossy fiber system (Blackstad et al., '70; Gaarskjaer, '78a,b), terminate in well defined laminae. The laminar organization of the hippocampal region can be visualized with Timm's sulfide silver method and has previously been used as a basis for comparison of more distantly related mammals (West and Andersen, '80; West et al., '84; West and Schwerdtfeger, '85). In addition, recent transplantation, x-irradiation, and lesioning studies have demonstrated a high degree of connective plasticity within the hippocampal region (Cook and Crutcher, '85; Cotman, '79; Laurberg and Zimmer, '80b; Zimmer et al., '86). Experimentally induced changes in the connectivity of the hippocampal region in part resemble differences found between species or strains and may involve mechanisms similar to those by which changes in hippocampal connectivity are effected during phylogeny (see for review, Finlay et al., '87). These results will be discussed with regard to the natural variations observed in the two *Apodemus* species.

MATERIALS AND METHODS

Animals and staining procedure

The animals used in this study were divided into three groups: the first group consisted of six *A. flavicollis* from Aarhus, Denmark (mean body weight 25 g, SD 1.1 g); the second group comprised six *A. sylvaticus* from Copenhagen, Denmark (mean body weight 17 g, SD 1.9 g); and the third group comprised six *A. sylvaticus* from Montpelier, France (mean body weight 21 g, SD 4.3 g). Only adult males were examined.

The brains of the animals were processed according to the Timm sulfide silver method (Timm, '58) as modified by Danscher and Zimmer ('78). The animals were anesthetized with Nembutal and perfused transcardially at 120 mmHg for 5 minutes with a buffered 1% sodium sulfide solution. The brains were removed quickly, frozen with gaseous CO_2 and cut in four parallel series of 30-μm-thick sections. All brains were cut in the horizontal plane. Two series from each brain were exposed to the physical development of the Timm stain, rinsed, dehydrated, and mounted with Dammar resin. Additional series were cell- or fiber-stained to verify the delineation of the hippocampal components.

The particular modification of the Timm stain used here was optimal for defining the sublayers of the molecular

layer of the area dentata. Other modifications of the Timm stain (Danscher, '81) and the selenium method (Danscher, '82) result in less background staining and specific definition of the catalytic sites (Danscher et al., '85; Pérez-Clausell and Danscher, '85) that is, zinc in synaptic vesicles. For comparison of the general appearance of the staining pattern, additional animals were treated according to these protocols.

Terminology and definitions

The terminology employed in this study is based on that used to describe the hippocampal region of the rat (Blackstad, '56; Haug, '74). The boundaries of the hippocampal components that were quantified have been described in detail previously (West et al., '78) and are illustrated in Figure 1. No distinction was made between the parts of the components of Ammon's horn belonging to regio superior (CA1) and regio inferior (CA3). The telencephalic borders encompassed the corpus callosum, striatum, and amygdala and excluded the septum, alveus, fornix, hippocampal commissures, and capsula interna.

Volumetric determinations

Volumes of the hippocampal components and of the telencephali were determined by using an unbiased technique based on the Archimedes-Cavalieri principle. Between 37 and 39 sections covering the entire dorsoventral extent of the hippocampal region were individually selected for each brain. The distance between the selected sections was 120 μm at dorsal and ventral levels, where the appearance of the hippocampal components changed markedly from section to section, and 240–360 μm in the middorsoventral parts, where profiles of the components appeared to change very little. The telencephalic volumes were calculated from evenly spaced sections taken at 240-μm intervals. Drawings of the borders of the components, as they appeared in Timm-stained sections, were made at a magnification of $\times$13 for the telencephalon, $\times$50 for the Ammon's horn and subiculum, and $\times$60 for the area dentata. The procedure for determining the volumes was the same as that previously used with other species (West and Andersen, '80; West et al., '78,'84; West and Schwerdtfeger, '85). In short, the drawings were digitized with a computer-controlled (PDP-11, DEC) Vidissector-camera (ITT). The volumes of the components were calculated from the areas of the profiles and the distance between the respective sections.

An allometric comparison, which was originally intended, was discarded because of difficulties both in choosing the appropriate exponent in the allometric equation, which varies greatly in different taxonomic groups of lower levels (Mann et al., '88), and in estimating correct body weights, the second crucial parameter of the allometric equation, which varies both with the geographic origin of animals and with the differential responses of wild animals to laboratory conditions. Instead, comparisons of the volumes of the components were based on ratios. The mean volume of a component in one of the groups was divided by the mean volume of the homologous component in one other group. Ratios for *A. flavicollis*, Denmark, to *A. sylvaticus*, Denmark, and *A. sylvaticus*, France, to *A. sylvaticus*, Denmark, will be presented. Ratios of the major subdivisions of the hippocampal region—that is, those for subiculum, Ammon's horn, and area dentata—were compared with that for the telencephalon, whereas ratios of the components of these structures were compared with ratios of the structure to which they belong. The significance of differences in ratios was determined with two-tailed *t*-tests. Several ratios were involved in more than one comparison. To avoid statistical inference by partly repeated comparisons, repetition was assumed to be complete and, according to Bonferoni *t*-statistics (Miller, '66), the P-value considered to be significant was divided by the number of comparisons; 19 comparisons were made, and P-values less than 0.0026 were considered significant.

RESULTS

Staining pattern

Timm staining patterns of the hippocampal regions of the two species were compared with that of the Timm-stained hippocampal region of the Wistar rat (Haug, '74). Only deviations from the normal staining pattern of the rat are noted here.

Haug pointed out that at ventral hippocampal levels the transition from the superficial zone of the dentate molecular layer to the intermediate zone was not as sharp as the transition from the intermediate to the deep zone. This pattern is reversed in the two *Apodemus* species. At ventral levels, the border between the superficial zone (lateral perforant path) and the intermediate zone (medial perforant path) is sharply delineated, whereas the intermediate zone is slightly stained at its midpoint along the radial axis. The density of staining gradually increases toward the deep (commissural-associational) zone of the dentate molecular layer. In both *Apodemus* species, the part of the stratum moleculare of Ammon's horn that appears to be a continuation of the superficial zone of the dentate molecular layer into regio inferior (CA3) of Ammon's horn does not become darker relative to the superficial zone as it does in the Wistar rat. Finally, the deep zone of the the dentate molecular layer of *A. flavicollis* was bilaminar with a pale outer zone and a dark inner zone (Fig. 2a,c). This bilaminar appearance started at the dentatohippocampal transition of the suprapyramidal blade of the area dentata and continued to the crest of the granule cell layer, where the pale outer zone gradually attained the appearance of the dark inner zone. Lamination was confined to the dorsal half of the area dentata and disappeared gradually at progressively more ventral levels. Along the transverse axis, the loss of bilaminar appearance occurred first in the region of the crest and later at the dentatohippocampal transition. This bilaminar deep zone is found neither in *A. sylvaticus* (Fig. 2b) nor in the Wistar rat.

Regional differences in the staining patterns of other hippocampal components of the rat were also found in the two *Apodemus* species and, with the exception of a diffuse staining between the somata of pyramidal and granule cells (Fig. 2a as compared with 2c), could be verified in sections stained with the Timm (Danscher, '81) and selenium methods (Danscher, '82). An example is the lamination of the deep zone of the dentate molecular layer in *A. flavicollis* (Fig. 2c). The quantitative measures, based on the modification of Timm's sulfide silver method used in this study, appear to reflect the properties of zinc-containing systems without interference from nonspecific staining.

Volumetric comparison

Table 1 lists mean volumes of the components examined and size ratios of homologous components. Volumes of the components lying in the left and right hemispheres were determined separately in the two Danish groups. Although

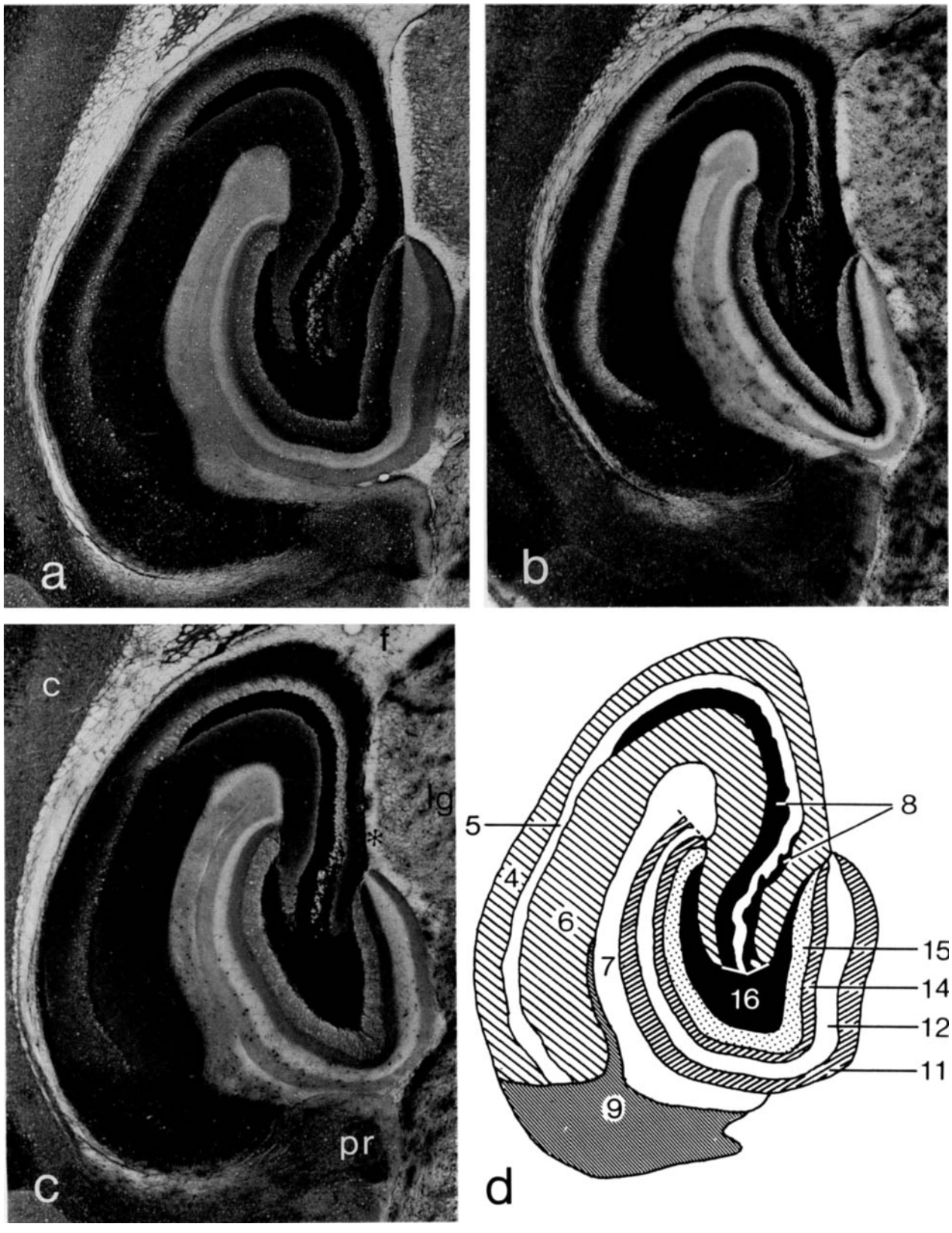


Fig. 1. Photomicrographs of the Timm-stained hippocampal region.**a:** *A. flavicollis*, Denmark; **b:** *A. sylvaticus*, Denmark; and **c:** *A. sylvaticus*, France; and **d:** A camera lucida drawing of the borders of the components measured (see Table 1), taken from a. Additional structures indicated in c. Ig, lateral geniculate body; f, fimbria; c, cerebral cortex; pr, presubiculum; asterisk, optic tract. ×32.

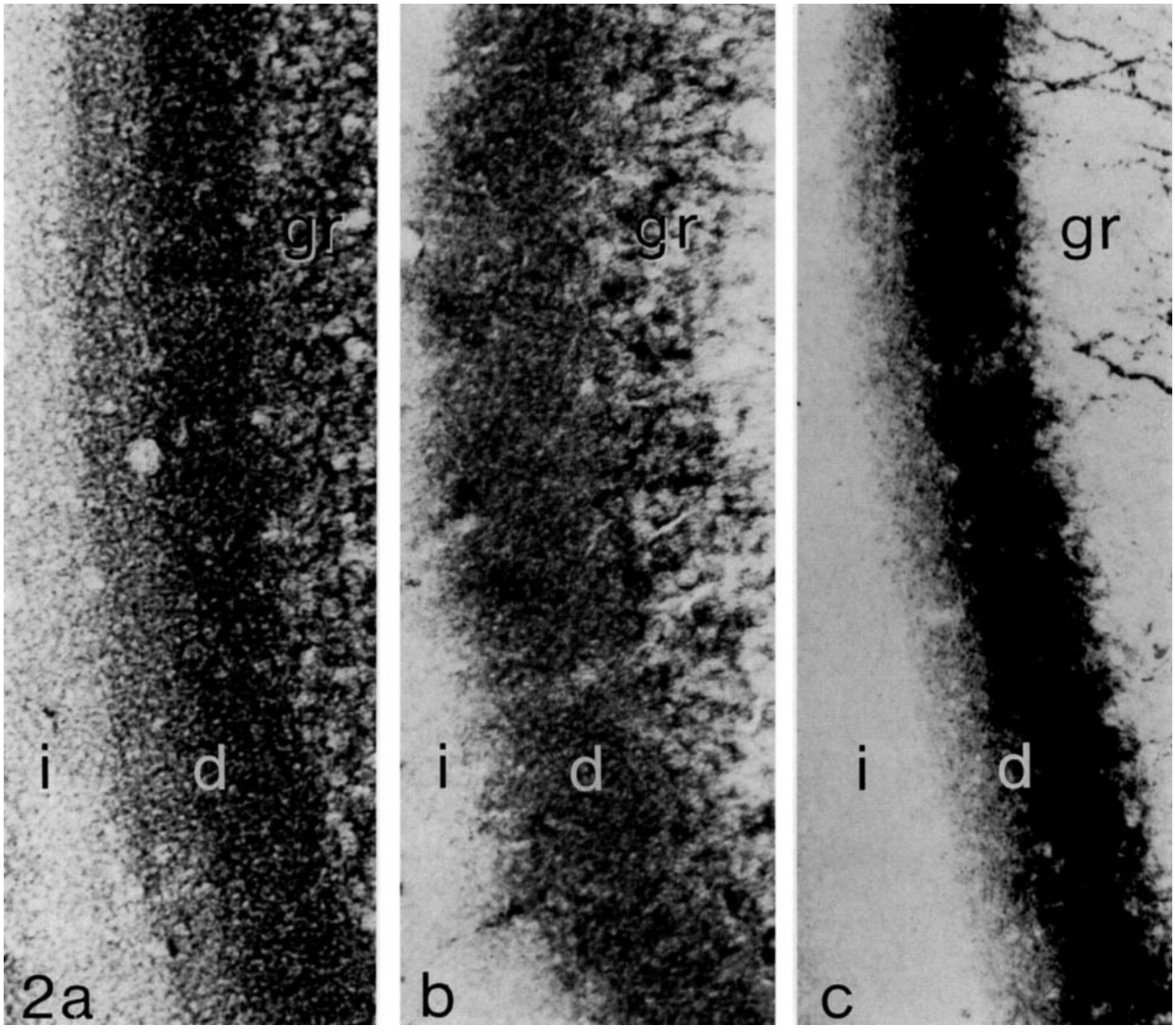


Fig. 2. Photomicrographs of the deep zone of the molecular layer of the suprapyramidal blade of the area dentata. **a:** *A. flavicollis*, Denmark; **b:** *A. sylvaticus*, Denmark. Both Timm-stained according to Danscher and Zimmer ('78). **c:** *A. flavicollis*, Denmark, stained according to the selenium method (Danscher, '82). i, intermediate, d, deep zone of the molecular layer; gr, granule cells. ×320.

a significant asymmetry was found in the deep subiculum of *A. flavicollis* (Slomianka and West, '87) the results of the interspecific comparison of the relative sizes of the deep subiculum were similar for both hemispheres. Consequently, in this study, only data obtained from one hemisphere will be presented: the left hemisphere, which was chosen arbitrarily. The *t*-values obtained from the comparisons of the size ratios are listed in Table 2.

Differences in the volume of Ammon's horn relative to the telencephalon were nonsignificant, both interspecifically and intraspecifically. The only interspecific difference found within the Ammon's horn was a significantly larger mossy fiber zone in *A. sylvaticus* than in *A. flavicollis*. Volumes of the other components relative to the volume of the entire Ammon's horn remained unchanged. Intraspecifically, that is, between the two groups of *A. sylvaticus*, all differences within the Ammon's horn were nonsignificant.

The subiculum is significantly larger in *A. flavicollis* than it is in *A. sylvaticus*, relative to both the volume of the telencephalon and the volume of Ammon's horn. No significant intraspecific differences could be found.

Interspecifically, the volume of the area dentata, relative to that of the telencephalon, was larger in *A. flavicollis* than in *A. sylvaticus*. Within the area dentata, the terminal field of the medial perforant path, that is, the intermediate zone of the molecular layer, was significantly smaller in *A. flavicollis* than in *A. sylvaticus*, whereas the hilus and the deep zone of the molecular layer were larger. One problem associated with comparison of the relative sizes of brain components is that the increase in the volume of one component must result in a relative decrease in the volumes of others. However, the above interpretation is validated by the localization of complementary change to only one other component, as no significant differences were found for the

TABLE 1. Volumes (mm^3) of Structures and Ratios of Volumes of the Two Group Comparisons[1]

	A. flavicollis DK		*A. sylvaticus* DK		*A. sylvaticus* F					
	mm^3	(S.D.)	mm^3	(S.D.)	mm^3	(S.D.)	R1	(S.E.)	R2	(S.E.)
1. Telencephalon	185.23	(16.44)	143.96	(5.64)	154.34	(9.88)	1.29	(0.05)	1.07	(0.03)
2. Ammon's Horn and area dentata	22.35	(1.68)	16.02	(1.35)	20.22	(1.88)	1.40	(0.06)	1.26	(0.07)
3. Ammon's Horn	15.69	(1.07)	11.54	(1.00)	14.22	(1.15)	1.36	(0.06)	1.23	(0.06)
4. Stratum oriens	4.31	(0.29)	3.04	(0.21)	3.86	(0.44)	1.42	(0.06)	1.27	(0.07)
5. Stratum pyramidale	2.22	(0.22)	1.59	(0.28)	1.82	(0.10)	1.40	(0.12)	1.15	(0.09)
6. Stratum radiatum	5.72	(0.34)	4.22	(0.28)	5.30	(0.47)	1.36	(0.05)	1.26	(0.06)
7. Stratum moleculare, Ammon's horn and subiculum	2.47	(0.29)	1.87	(0.21)	2.19	(0.13)	1.32	(0.09)	1.17	(0.06)
8. Mossy fiber zone	0.97	(0.07)	0.84	(0.08)	1.06	(0.17)	1.16	(0.06)	1.27	(0.10)
9. Deep subiculum	4.62	(0.54)	2.76	(0.23)	3.15	(0.22)	1.68	(0.10)	1.14	(0.05)
10. Area dentata	6.67	(0.69)	4.48	(0.41)	6.16	(1.08)	1.49	(0.08)	1.37	(0.11)
11. Stratum moleculare, superficial zone	1.92	(0.19)	1.23	(0.10)	1.56	(0.31)	1.57	(0.08)	1.27	(0.11)
12. Stratum moleculare, intermediate zone	1.32	(0.16)	1.11	(0.12)	1.46	(0.26)	1.18	(0.08)	1.31	(0.11)
13. Combined superficial and intermediate zones	3.24	(0.31)	2.34	(0.21)	3.02	(0.56)	1.38	(0.07)	1.29	(0.11)
14. Stratum moleculare deep zone	1.05	(0.11)	0.62	(0.08)	0.94	(0.18)	1.70	(0.11)	1.53	(0.14)
15. Stratum granulosum	1.24	(0.21)	0.88	(0.12)	1.21	(0.21)	1.42	(0.12)	1.38	(0.12)
16. Hilus	1.14	(0.10)	0.65	(0.05)	0.99	(0.16)	1.77	(0.09)	1.53	(0.12)
17. Combined hilus and mossy fiber zone	2.13	(0.10)	1.55	(0.10)	2.06	(0.33)	1.37	(0.04)	1.33	(0.09)

[1](R1 = *A. flavicollis*, Denmark/*A. sylvaticus*, Denmark, R2, *A. sylvaticus*, France/*A. sylvaticus*, Denmark). n = 6 for all groups.

superficial zone of the molecular layer and the granule cell layer.

Intraspecifically, the volume of the area dentata, relative to the telencephalon, was larger in the French *A. sylvaticus* group than in the Danish group. Within the area dentata, the terminal field of the lateral and medial perforant path, that is, the combined superficial and intermediate zone of the molecular layer, was smaller in the Danish *A. sylvaticus* group. No other intraspecific differences were observed.

Because the mossy fiber zone of the Ammon's horn and the hilus in combination represent the terminal field of the granule cells of the area dentata, these two structures were pooled (structure 17 in Table 1 and 2) and their cumulative volume compared with both the volume of the Ammon's horn and the area dentata. Differences in the relative sizes of the hilus and mossy fiber zone observed in the interspecific comparison complemented each other, with the result that no significant interspecific differences were found in their combined volumes in either comparison.

Recently, it has been shown that the extent of terminal fields visualized with Timm's sulfide silver method may either determine or covary with the extent of terminal fields of other projections, which have the same laminar borders, such as the terminal field of the septohippocampal projection (Zimmer et al., '86). Differences in the relative sizes of the terminal fields of these other systems may be similar to differences observed in the Timm-stained hippocampal components. Their discussion, however, lies beyond the scope of this study.

TABLE 2. Two-tailed *t*-Test for the Relative Size of the Hippocampal Components: *t*-Values[1]

	A. flavicollis Dk vs. *A. sylvaticus* Dk	*A. sylvaticus F* vs. *A. sylvaticus* Dk
1 vs. 2	3.4279	4.3940*
1 vs. 3	2.1496	3.8412
1 vs. 9	8.1801**	2.3032
1 vs. 10	4.3572*	4.0128*
3 vs. 4	2.1981	1.1752
3 vs. 5	0.8186	1.4693
3 vs. 6	−0.0934	1.0563
3 vs. 7	−1.0504	−2.1340
3 vs. 8	−5.4699**	0.5521
3 vs. 9	5.2102**	−2.6125
3 vs. 10	2.7803	2.1119
3 vs. 17	0.1139	1.2378
10 vs. 11	2.0410	−2.9812
10 vs. 12	−8.3981**	−2.3569
10 vs. 13	−4.4207**	−4.2263*
10 vs. 14	4.1306*	3.3914
10 vs. 15	−1.5717	0.1992
10 vs. 16	7.5474**	3.7641
10 vs. 17	1.9429	−0.9527

[1]The numbers of the ratios compared correspond to the numbers of structures and their ratios in Table 1.
*$0.0026 > P > 0.0005$.
**$0.0005 > P$

DISCUSSION

The three major subdivisions of the hippocampal region differ with regard to the amount of interspecific and intraspecific variability. These differences provide an opportunity to discuss the apparent stability of the relative volumes of the components of Ammon's horn in relation to phylogenetic trends beyond the species level and an opportunity to discuss the possible structural changes associated with speciation, which may account for the differences observed in the relative volumes of the components of the area dentata. Changes due to speciation are considered to be related to phylogenetic development and consequently may be propagated as phylogenetic trends. Because this continuous process cannot be observed directly, knowledge of expected variation at different taxonomic levels facilitates identification of both phylogenetic trends and the brain regions in which variations may relate more to ecology than to taxonomy (Mace et al., '81; Mann et al., '88). The findings

presented in this study provide a minimum estimate of what can be expected in interspecific and intraspecific differences in the hippocampal components.

The deep subiculum

The volume of the deep subiculum relative to that of the telencephalon and that of Ammon's horn is significantly different in the two species examined, whereas it is similar in the spatially separated groups of *A. sylvaticus*. Based on allometric comparisons, the volume of the deep subiculum relative to body weight has previously been shown to have a poor correlation with the volume of Ammon's horn and its components (West and Schwerdtfeger, '85). Considerable variation in the volume of the deep subiculum within species belonging to the same taxonomic unit, as found in this study, may explain this observation.

Ammon's horn

Data presented in this study indicate that Ammon's horn is well suited for comparative studies of distantly related taxonomic groups. No significant differences were found in the volume of Ammon's horn relative to the volume of the telencephalon in the two closely related species or in spatially separated groups of the same species. With the exception of the mossy fiber zone, this is also true for components of the Ammon's horn. A minimum estimate of no change on the basis of only two species is of limited value. However, evidence that Ammon's horn has limited variability can be found in the literature. First, Stephan et al. ('74) noted the restricted variation in the hippocampus, the main part of which is Ammon's horn, in comparison of Megachiroptera and Microchiroptera. Second, when the allometric size indices of regio superior of Ammon's horn are compared with those of the hippocampus, the variation observed in the hippocampus is almost twice the variation observed in components of the regio superior of Ammon's horn alone (Stephan, '83). Finally, the slight variation is in agreement with the idea of a past size-stabilizing selection of the components of Ammon's horn as proposed by Crusio et al. ('86) on the basis of the results of a diallel cross of different inbred strains of mice. Changes in the size of these components observed in earlier studies—i.e., a larger stratum radiatum and smaller stratum oriens in the rat as compared with the European hedgehog (West et al., '84) and an increase, from the hedgehog to the tree shrew and marmoset, in the size of Ammon's horn, due mainly to an increase in the size of the components that are part of regio superior (West and Schwerdtfeger, '85)—are likely to be related to trends in the phylogenetic development of the Ammon's horn, from insectivores to rodents and primates, respectively. The mossy fiber zone, together with the hilus of the area dentata, is discussed below.

On the basis of allometric comparisons, an increase in the size of the Ammon's horn, resulting primarily from an increase in the size of the regio superior (CA1), has been established as a stable trend in the phylogenetic development of the brain, from basal insectivores to primates (Stephan, '83; West and Schwerdtfeger, '85). Although the amount of progression appears to be different for the regio superior and regio inferior, both structures increase in size during phylogenetic development. From these studies, it can be expected that a change in the relative size of a component in regio superior will be accompanied by a change in the size of the corresponding component in regio inferior, that is, smaller in magnitude but in the same direction. As a consequence, the lack of information in this study about the relative sizes of the components of Ammon's horn lying in the regio inferior and the regio superior does not impose significant limitations on the detection of interspecific and intraspecific differences.

Area dentata

Hilus and deep zone of the molecular layer. Within the area dentata, the hilus and the deep zone of the molecular layer are the components with the largest interspecific differences. The similar differences in volumes of the hilus and the deep zone of the stratum moleculare of the area dentata may be related to the anatomic connection of these two zones in that the polymorph layer of the hilus is known to project to the deep zone of the dentate molecular layer (Berger et al., '81; Hjorth-Simonsen and Laurberg, '77; Laurberg, '79). Similar correlations in the sizes of these two components have been observed in other rodents (West and Andersen, '80) and in distantly related species (West and Schwerdtfeger, '85).

The axons of the dentate granule cells terminate in the hilus and in the mossy fiber zone of Ammon's horn. The lack of significant differences in the combined volumes of the hilus and the mossy fiber zone in the two species studied here is compatible with the lack of differences observed in the relative sizes of the granule cell layers. Because the silver precipitate of the Timm stain is located within the mossy fiber boutons (Haug, '67; Ibata and Otsuka, '69; Pérez-Clausell and Danscher, '85) each unit of granule cell volume appears to give rise to a unit volume of mossy fibers boutons. The total volume of the terminal fields of the granule cell axons are identical proportionally in the groups studied but are distributed in different hippocampal regions in the different species. Earlier studies have shown that the mode of termination of the mossy fibers in the hippocampal region is highly variable both within strains of a species (mice: Barber et al., '74; Heimrich et al., '85; cats: Laurberg and Zimmer, '80a) as well as between different species (rat: Blackstad et al., '70; Gaarskjaer '78a,b; hedgehog: Gaarskjaer et al., '82; guinea pig: Geneser-Jensen et al., '74). After having been experimentally deprived of their normal target areas during development, the dentate granule cells have been shown to redistribute their axons within the Ammon's horn (Cook and Crutcher, '85; Laurberg and Zimmer, '80b). In addition, changes in the relative sizes of the terminal area normally occupied by the collaterals and the main axons of the granule cells can be found in dentate transplants, in which very few hippocampal pyramidal cells are attached to the transplant. In these preparations there is an extensive arborization of mossy fibers within the part of the transplant that is homologous to the hilus (Sunde and Zimmer, '83). In summary, there appears to have been a redistribution of mossy fiber boutons during the phylogenetic development of the two species studied here, which involved changes in the relative amounts of boutons associated with mossy fiber collaterals within the hilus and complementary changes in the relative amounts of boutons associated with the main mossy fiber axons within Ammon's horn.

Perforant path. Afferents from the medial and lateral entorhinal cortices to Ammon's horn arise from pyramidal

cells in layer III and terminate within the stratum moleculare of regio superior of Ammon's horn and the subiculum. Stellate cells within layer II of the medial and lateral entorhinal cortices give rise to the perforant path, terminating in the intermediate and superficial zones of the dentate molecular layer. In addition, they innervate the regio inferior of Ammon's horn (Steward, '76; Steward and Scoville, '76).

The presence of an interspecific difference in the relative size of the intermediate zone and the lack of differences in the superficial zone of the dentate molecular layer indicate that the size of the terminal fields of the stellate cells of the medial and lateral entorhinal cortex can change independently. In addition, changes in the sizes of the terminal fields of the entorhinal layer II stellate cells appear to have taken place independently of changes in the size of the terminal fields of entorhinal layer III pyramids, because differences in the combined volumes of the superficial and intermediate zones of the dentate molecular layer observed in the intraspecific comparison were not accompanied by differences in the relative size of the terminal fields of the afferents from entorhinal layer III pyramids within the regio superior and the subiculum, which constitute the major part of the stratum moleculare of Ammon's horn and the subiculum.

Recent studies have demonstrated that granule cells can expand their receptive fields if a surplus of entorhinal input is provided (Poulsen et al., '87). Similar shifts in the size of the receptive field of the dentate granule cells may permit an increase in combined relative volumes of the superficial and intermediate zones of the dentate molecular layer in French *A. sylvaticus* as compared with Danish *A. sylvaticus*.

Another mechanism for providing the additional postsynaptic space associated with a larger perforant path input may be a shift in the availability of dendritic sites from hilar commissural-associational afferents to perforant path afferents. The relatively smaller deep zone of the dentate molecular layer and the relatively larger size of the intermediate zone of the dentate molecular layer in *A. sylvaticus* in comparison to those of *A. flavicollis* are compatible with this explanation. Although a corresponding change in the size of the entorhinal terminal fields cannot be induced experimentally by partial or complete denervation of the deep zone of the dentate molecular layer (Cotman, '79), this restriction may not exist during phylogenetic development.

CONCLUSIONS

Of the three subdivisions of the hippocampal region studied, Ammon's horn appears to be most appropriate for comparative quantitative studies of distantly related species, as virtually no differences could be observed in the relative size of the components of Ammon's horn in the two closely related species described here. The exception to this were the separate zones of mossy fibers found in the regio inferior of Ammon's horn and the hilus. Significant differences in the relative size of the components could be observed in the area dentata and the subiculum. These components should consequently be treated with care when used in comparative quantitative studies aimed at revealing phylogenetic trends in this region of the brain. The inclusion of several species from each taxonomic unit would reduce interference from differences in the size of these components that might be specific to a particular species.

Morphologic changes associated with variations in the relative sizes of the hippocampal region observed here are compatible with those expected in closely related species (Ebbeson, '80; Ulinski, '80), that is, changes in the size of specific afferent and efferent connections, and, as a consequence, rearrangements in the axonal and dendritic organization of specific hippocampal components. Morphologic changes of these types that can explain the natural variation in the quantitative composition of the hippocampal region can be induced experimentally during ontogenetic development.

Differences in the relative proportions of the components of the hippocampal region observed in this study are confined to 1) the subiculum, which is a nodal point of extrinsic hippocampal afferent and efferent connections (Swanson et al., '82); 2) the area dentata, which is the point of entrance to the unidirectional chain of intrahippocampal connections (Schwerdtfeger, '84; Swanson et al., '82); and 3) the mossy fibers, which channel information flow to Ammon's horn and modulate this flow by way of the reciprocal connection of the hilus with the dentate granule cells. As is evident from the description of the Timm staining pattern of the dentate molecular layer of these and other species (Fredens, '81; Stengaard-Pedersen et al., '83), comparative studies of cholecystokinin and enkephalin immunoreactivities (Fredens et al., '87; Stengaard-Pedersen et al., '83), the acetylcholinesterase chemoarchitecture (Geneser, '87), and septotemporal termination patterns and densities of the intrinsic commissural connections of the hippocampus (Van Groen and Wyss, '88), a considerable amount of qualitative differences is found particularly within the area dentata of different species. Demands on hippocampal information processing specific to a species appear to be accommodated by changes in the relative size and the qualitative properties of the afferent and efferent connections of the subiculum and the area dentata, whereas components of Ammon's horn change very little both qualitatively and quantitatively.

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